J Braz. Soc. Mech. Sci. Eng. (2017) 39:1929–1945
DOI 10.1007/s40430-017-0739-3

TECHNICAL PAPER

Energy saving of a hydrostatic drive system by incorporating soft switch

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Received: 23 September 2016 / Accepted: 17 February 2017 / Published online: 1 March 2017
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Abstract A soft switching concept for the purpose of energy saving during on–off valve controlling of a hydrostatic drive system has been reported in this article. The soft switch has been incorporated in a state-of-the-art configuration in the drive circuit. Some identified soft switch parameters have been optimized with respect to minimum throttling energy loss using multi-run-simulation in bond graph. The bond graph technique has been used for the simulation study as it is based on power flow representation and guarantees energetic consistency in the model structure. Two different bond graph models, with and without the soft switch, have been simulated and validated experimentally; and the power and energy losses during spool movements of the directional control valve have been compared to quantify the energy savings. The study reveals that savings of more than 50% throttling energy can be possible using a soft switch in a switched mode hydrostatic drive system.

Keywords Bond graphs · Energy loss · Hydrostatic transmission system (HST) · Regression · Soft switch

List of symbols

a_0, a_1, a_2, a_3, and a_4	Constant coefficients
A_{check}	Port opening area of the orifice of the check valve (m^2)
A_{lv}	Port opening area of the orifice of the unloading DC valve (m^2)
A_{port}	Port opening area of the port of the soft switch (m^2)
A_{P_x}	Port opening area of the DCV (m^2)
A_{ss}	Area of the piston of the soft switch (m^2)
A_v	Valve orifice area (m^2)
C	Single port energy storage capacitor element in bond graph model
C_d	Coefficient of discharge
C_D	Coefficient of flow through check valve
C_{ss}	Radial clearance of the piston of the soft switch (μm)
De	Effort dictator element in bond graph model
Df	Flow dictator element in bond graph model
D_m	Volume displacement rate of the hydro-motor (m^3 rad^{-1})
D_p	Volume displacement rate of the radial piston pump (m^3 rad^{-1})
e	Effort in bond graph model
$E_{throtloss}$	Throttling energy loss (J)
f	Flow in bond graph model
$F_{preload}$	Preload of the spring of the soft switch (N)
I	Single port energy storage inertial element in bond graph model
J	Load inertia of the rotating shaft of the hydro-motor (kg m^2)
K	Bulk modulus of air free flowing fluid (N m^{-2})

Technical Editor: Kátia Lucchesi Cavalca Dedini.

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k_{check}	Flow constant of the check valve, $k_{check}(= C_D A_{check}\sqrt{2/\rho})$ ($kg^{-0.5}\ m^{3.5}$)
K_e	Effective bulk stiffness of the fluid ($N\ m^{-2}$)
k_f	Valve flow constant, $k_f\left(= C_d\sqrt{2/\rho}\right)$ ($kg^{-0.5}\ m^{1.5}$)
k_l	Flow constant of the unloading DC valve ($kg^{-0.5}\ m^{1.5}$)
K_{pm}	Bulk stiffness of the fluid ($N\ m^{-5}$)
K_{port}	Flow constant of the port of the soft switch ($kg^{-0.5}\ m^{1.5}$)
K_{ss}	Spring stiffness of the soft switch ($N\ m^{-1}$)
l_{ss}	Piston length of the soft switch (m)
l_{ss_port}	Axial length of the port of the soft switch (m)
M	Number of samples
M_{ss}	Piston mass of the soft switch (kg)
P_0	Atmospheric pressure (bar)
P_{crack}	Cracking pressure of the check valve, bar
P_i	Precision error in each variables
P_{mi}	Pressures at the inlet (upstream side) of the hydro-motor (bar)
P_{mo}	Pressures at the outlet (upstream side) of the hydro-motor (bar)
P_{pp}	Loading pump plenum pressure (bar)
P_{rear}	Pressure at the rear end of the soft switch (bar)
P_{switch}	Switched volume pressure (bar)
P_{tank}	Tank pressure (bar)
$P_{Pthrotloss}$	Precision limit of the throttling power loss (W)
$P_{Pswitch}$	Precision error of switched pressure (bar)
P_{Pmi}	Precision error of upstream pressure (bar)
P_{Pmo}	Precision error of downstream pressure (bar)
$P_{\omega m}$	Precision error of angular speed ($rad\ s^{-1}$)
Q_{chv_tb}	Flow rate through the check valve (tank to the rear end of the soft switch) ($m^3\ s^{-1}$)
Q_{chv_ip}	Flow rate through the check valve (soft switch to the mainstream) ($m^3\ s^{-1}$)
Q_{dcvP_x}	Flow rate through 4/2 DCV port ($m^3\ s^{-1}$)
Q_{supply}	Flow supply from fixed displacement pump ($m^3\ s^{-1}$)
Q_{ss_port}	Flow rate through the port of the soft switch to the tank ($m^3\ s^{-1}$)
R	Single port resistive element in bond graph model
R_a	Volume fraction of entrained air in the oil
R_{bl1}, R_{bl2}	Pipe friction resistance at upstream and downstream sides, respectively ($N\ s\ m\ rad^{-1}$)
R_f	Viscous friction resistance ($N\ s\ m\ rad^{-1}$)
R_{high}, R_{low}	External leakage resistances at the upstream and the downstream side of the system, respectively ($N\ s\ m^{-5}$)
R_{ipl}	Cross-port leakage resistance of the hydro-motor ($N\ s\ m^{-5}$)
R_{lkg1}, R_{lkg2}	External leakage resistances at the inlet and the outlet ports of the hydro-motor, respectively ($N\ s\ m^{-5}$)
R_{sp}	Friction resistance between the piston and the soft switch wall
r_{ss}	Piston radius of the soft switch (m)
SD	Standard deviation
Se	Single port *source of effort* element in bond graph model
Sf	Single port *source of flow* element in bond graph model
S_i	Bias error in each variables
$S_{Pthrotloss}$	Bias limit of the throttling power loss (W)
$S_{Pswitch}$	Bias error of switched pressure (bar)
S_{Pmi}	Bias error of upstream pressure (bar)
S_{Pmo}	Bias error of downstream pressure (bar)
$S_{\omega m}$	Bias error of angular speed ($rad\ s^{-1}$)
TF	Transformer element in bond graph model
t_{factor}	Coverage factor
$U_{Pthrotloss}$	Total experimental uncertainty (W)
V_{ss_clear}	Clearance volume at the rear end of the soft switch (m^3)
$\dot{V}_1$	Flow passes through the direction control valve (DCV) ($m^3\ s^{-1}$)
$\dot{V}_2$	Flow returns to the tank ($m^3\ s^{-1}$)
v_{ss}	Velocity of the soft switch ($m\ s^{-1}$)
V_{switch}	Switched volume (m^3)
x_{ss}	Instantaneous position of the soft switch (m)
x_{ss_crack}	Crack length of the piston travel (m)

x_{ss_max}	Maximum possible travel length (m)
ω_m	Angular speed of the hydro-motor (rad s^{-1})
ρ	Mass density of the oil (kg m^{-3})
γ	Ratio of specific heat for air
μ	Dynamic viscosity of the oil (N s m^{-2})
·	Time derivative of the variable
1	Junction element in bond graph model
0	Junction element in bond graph model

1 Introduction

The research on minimization of transportation energy recently has drawn much attention to reduce fuel consumption, greenhouse gas emissions and air pollutants. This article is concerned with energy savings for off-road vehicles (i.e., backhoe loaders, hydraulic excavators, rear-end dumpers, cement mixers, etc.), which mostly deal with the transport of raw bulk materials from one point to another while standing at a fixed location or moving on a rough terrain. For these applications, most of the power consumption takes place due to movement of the associated hydraulic actuators (i.e., cylinders and hydro-motors).

The off-road vehicles are driven by power hydraulics that deals with generation, control and transmission of power using pressurized fluids. The hydraulic oil provides a very rigid medium for transmitting power and thus can operate under high pressure to provide huge forces and torques to drive loads with utmost accuracy. The prime mover used for hydrostatic transmission system is mostly internal combustion engine, particularly in the case of pay loaders, excavators, and other construction equipment, wherefrom power is transmitted to the hydraulic drive through a power take-off.

The strategy of energy saving of a hydraulic system can be classified in three categories: electrical system by the use of batteries [1–3], mechanical system by means of flywheels [4], and hydraulic system using accumulators [5, 6]. Excluding these, fuel consumption can be reduced using a hybrid system [7, 8] and minimizing different losses, such as throttling, leakage, hysteresis, and compressibility losses, which may be attempted by introducing additional compliances to the system or energy regeneration [9].

In off-road vehicles, hydraulic systems frequently use on/off controlled valves to fulfill the operational requirements such as lifting and lowering of the boom and arm, tilting of the bucket, and slewing action of the deck. As a result, a significant amount of energy is lost during the transition of the high speed valves. In a hydraulic circuit, throttling energy losses during the valve transition are the major energy loss (60% of the total loss, approximately) in comparison to other losses (i.e., compressibility losses, hysteresis losses, etc.). Therefore, reducing throttling energy loss means saving a massive amount of energy from the waste energy of the system.

Tu et al. [10] proposed replacing the traditional on/off valve by a pulse width modulated (PWM) valve with unidirectional rotary spool. This concept [10] was experimentally validated by the same authors in [11]. They also characterized the primary losses of the rotary valve. In [12], Rannow and Li reported a novel idea to reduce significant amount of throttling energy loss during the valve transition by providing an alternative temporary path to store the extra amount of fluid when the valve is not in the fully opened condition. This mechanism is called soft switching and the device used to store the fluid is called the soft switch. This soft switch is an assembly of a piston, one compression spring and an externally operated locking and releasing mechanism. Initially, the piston is locked at the middle position of the soft switch and it is unlocked when the tank valve starts moving. The main challenge of that approach [12] was to design the locking and releasing mechanism with precise timing. The locking mechanism should be able to rapidly release the piston under a heavy loading pressure. The performance of the soft switch would depend upon this locking–unlocking mechanism, but there was no practical direction given for that in [12]. This drawback was addressed in an article authored by Ven De Ven [13], where a novel practical approach was reported using the pressure signal in the switched volume to trigger the releasing operation of the lock. Later on, Beckstrand and Van De Ven [14] experimentally validated the same theory.

All these studies were based on basic hydraulic circuits which contain a hydraulic pump, one directional control valve (DCV) and a constant load provided by a high pressure accumulator. But the performance of the soft switch in a typical hydraulic circuit with pump loading was not deliberated. Therefore, the soft switching concept should be tested in a typical hydraulic system to increase the fidelity in application. In this work, a hydro-static transmission (HST) system with pump loading, quite similar to that described by Dasgupta et al. [15], is chosen as a system to examine the performance of the soft switch. The bond graph technique for simulation has been used in this paper. A portrayal representation of any physical dynamic system can be done using bond graph [16–22]. It involves transforming the system into a dynamic simulation model in source code form and it is popularly used for analyzing systems having the interaction of multi energy domains.

In this work, the aforesaid soft switch integrated to an HST system in a state-of-the-art configuration has been reported aiming to reduce the power consumption in each throttling operation of the directional control valve (DCV).

Two models, with and without soft switch, have been simulated and validated experimentally. The throttling energy losses and power losses are compared for both the systems. The optimum soft switch parameter values have been obtained through multi-run simulation aiming to minimize the throttling energy loss. Finally, a relationship among the critical parameters of the soft switch has been derived from regression analysis.

2 The test set-up

The schematic of the proposed hydrostatic drive system with soft switch is shown in Fig. 1. A soft switch assembly (item 15) is mounted in a state-of-the-art configuration in the hydraulic test bed. The soft switch is an assembly of a specially designed annular grooved piston, compression spring, one cylinder, and two check valves. The function of the soft switch is to store an extra amount of throttle flow into the chamber of the cylinder during the high speed on/off valve transition and release it when the valve is completely open. In this system, a constant flow (Q_{supply}) is supplied from a fixed displacement axial piston pump (item 02).

To achieve the speed reversal, the constant flow (Q_{supply}) is passed through the 4/2 DCV (item 08) to drive the hydraulic motor (item 10). In contrast to the previous study [15], the 4/3 DCV of the hydraulic circuit is replaced by a 4/2 DCV to observe the instantaneous speed reversal. This leads to the development of a high pressure rise during the fast transition of the valve spool, which can be smoothed by the use of the soft switch and at the same time a significant amount of throttling energy loss can be arrested. The hydro-motor (item 10) is coupled to a radial piston pump (item 12). There is a provision for variation of pressure in the loading circuit by allowing fluid to flow through an unloading (or bypass) direction control valve (item 14), which is installed in parallel to a pressure relief valve (item 6). The bypass valve is solenoid operated and it is normally off. The soft switch is connected to the switched volume. The soft switch accessories, i.e., the two check valves at the rear end of the soft switch, are connected with the tank and the hydro-motor, respectively. The second check valve is used for safety purpose and also to supply the waste flow to the hydro-motor as a regenerative model. It may also help to reduce the throttling loss of the system. It may be noted that if the soft switch part is removed from

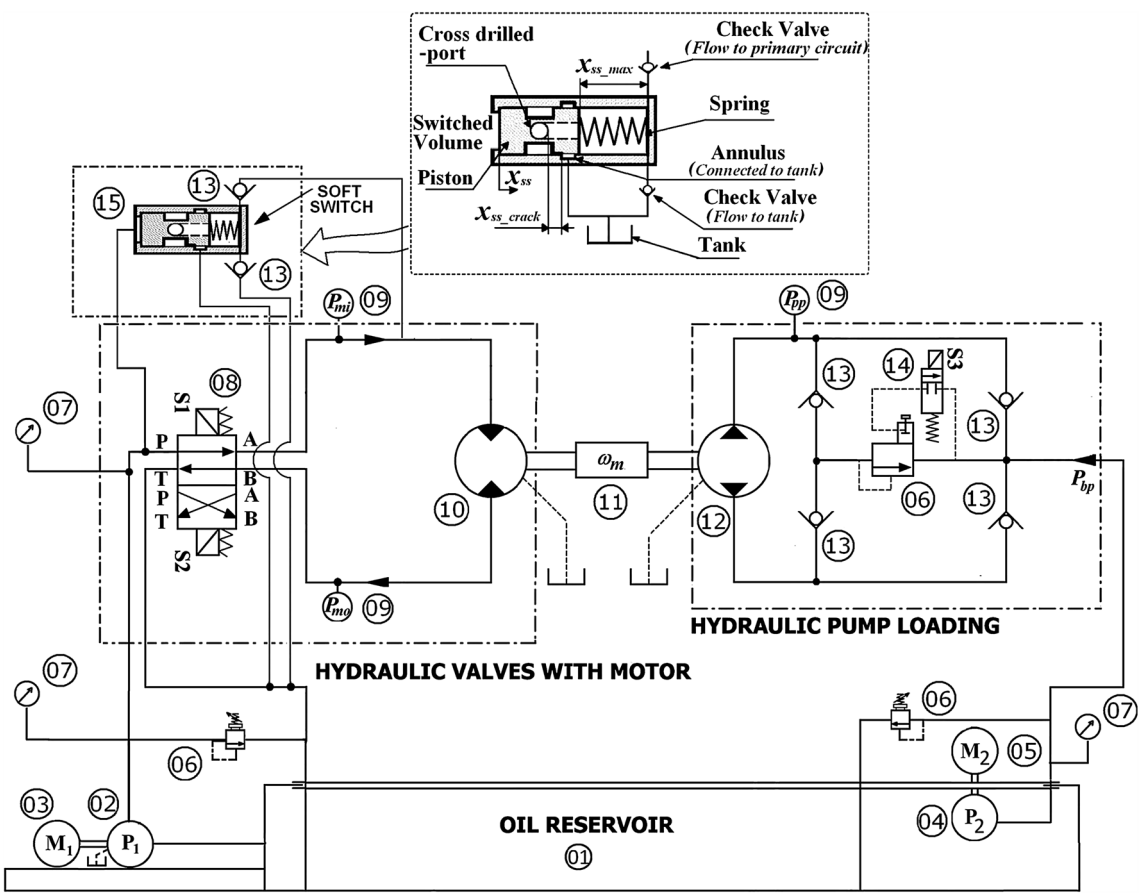

Fig. 1 An open-loop hydrostatic transmission system with soft switch

Fig. 1, the system is converted into the first system (i.e., HST system without soft switch). The list of materials of the system is given in Table 1.

After a certain time of fluid flow through the P-A port of the 4/2 DCV, the S_2 solenoid is activated, as a result the orifice of the P-A port of the valve begins to close and an instantaneous high pressure on the switched volume is developed. This causes an unbalanced force on the piston of the soft switch. This unbalanced force pushes the piston towards the rear end of the soft switch. After the complete closure of the P-A port and before opening of the P-B port, the pressure of the switched volume increases to its maximum. When the P-B port is completely opened, the pressure in the switched volume decreases slightly and reaches the steady state condition. In the next operation, when S_1 solenoid is energized, the P-B port starts to close and the pressure in the switched volume reaches its maximum value after the full closure of the P-B port and just before the opening of the P-A port. At the same time the piston of the soft switch move enough distance to open the internal passage of the soft switch. As a result, the pressure at the rear end of the soft switch decreases and it allows the fluid to flow through the check valve from the tank to the soft switch. Then, the piston returns to its original position and is ready for the next cycle after the complete opening of the P-A port.

The sequence of operations of the 4/2 DCV is shown in Fig. 2. After a particular period of the full opening, the P-A port starts closing at time t_1 and is completely closed at time t_2. Next, the P-B port starts opening at t_3 and is completely opened at t_4. After a certain dwell period (from t_4 to t_5), the P-B port starts closing at t_5 and is completely closed at t_6. In the next operation, the P-A port is opened (from t_7 to t_8). Thus, there are four transition periods in each cycle. The same process is repeated for the next cycle. In this study, two complete cycles are considered. Hence, the total number of transition events is eight.

3 Modelling of the system

The mathematical modelling of the system, both with and without the soft switch, has been developed using bond graphs. Bond graph is a powerful tool for modelling complex systems having nonlinearity and interaction with several energy domains. It also ensures energetic consistency in the model structure [12–18], which is important here because this study involves improvement of energy efficiency. The assumptions made in the modelling are the following:

1. The fluid inertia is neglected.
2. At the inlet port of the valve, a constant flow source is maintained.
3. The soft switch ports and valve orifices are considered rectangular.
4. The flow force effect is not taken into consideration.

Table 1 List of materials

S. no.	Name	Capacity	Make/model
1.	Tank	100 L	SPHL
2.	Fixed displacement axial piston pump	15 LPM (12 cc/rev)	HAWE
3.	Electrical motor	5 kW (7.5 HP), 1450 rpm, 420 V	LHP
4.	Booster pump (Gear)	7 LPM (5 cc/rev)	DOWTY
5.	Electrical motor	0.75 kW(1HP), 1450 rpm, 420 V	LHP
6.	Pressure relief valve	DPRH 06 S 315	Polyhydron
7.	Pressure gauge	0–280 bar, G FILLED, 1/4 BSP CONNECTION, 4″ DIAL	WIKA
8.	4/2 DCV	350 bar, 60 LPM	Rexroth, 4WE 6 C 6X/E G24 N9K4
9.	Pressure transmitter	S10–250, 4–20 ma, 0–200 bar	WIKA
10.	Fixed displacement hydraulic motor	Displacement—8.2 cm^3/rev	M + S Hydraulics
11.	Coupling		
12.	Radial piston pump	Displacement—5.31 cm^3/rev	Polyhydron
13.	Check valve	C 06 S1 03	Polyhydron
14.	Unloading DC valve	350 bar, 48 LPM	Polyhydron, 4DE 06D DSA220-01-3
15.	Soft switch	–	Fabricated
16.	Oil Type	Servosystem 68	Indian oil
17.	Speed sensor	12 mm Φ; 5–12 V; 100 mA	IPL
18.	Portable data recorder	–	Hydac Electronics HMG 3010

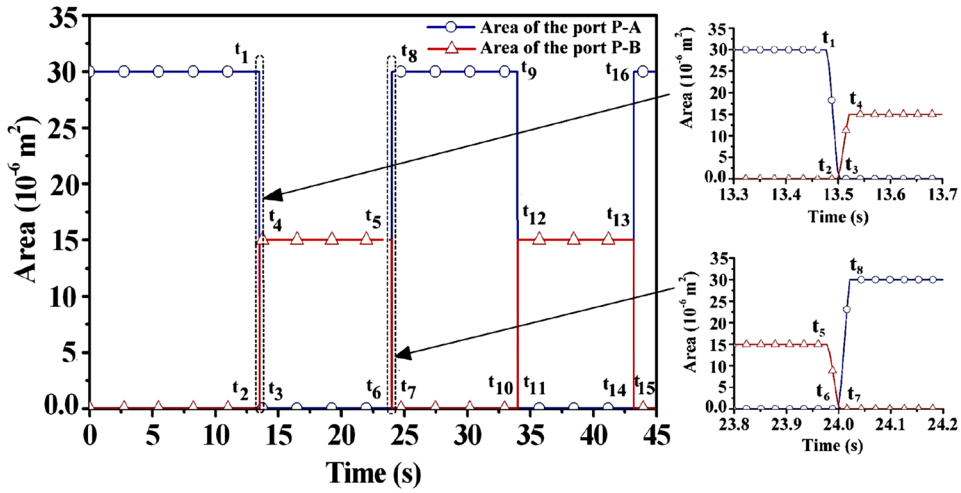

Fig. 2 Orifice area of 4/2 DCV for two cycles

5. The flow is considered isothermal.

3.1 Bond graph model of the system without soft switch

The bond graph model without soft switch is given in Fig. 3, where the constant flow supplies (Q_{supply}) from a fixed displacement bent axis pump is represented by Sf_1 element. The flow $\dot{V}_1$ through the direction control valve (DCV) drives the gerotor motor and the outflow V_2.returns to the tank through the DCV. The resistances of the fluid flow through the DCV are modelled by R_4 and R_{26} elements, which are nonlinear functions of the valve port opening areas. The P_{mi} and P_{mo} are the measured pressures at the inlet (upstream side) and outlet (downstream side) ports of the hydro-motor. The P_{switch} is the pressure of the switched volume, measured by the detector De_3. The C elements, i.e., C_7 and C_{23} represent the bulk stiffness of the fluid (K_{pm}) in the gerotor motor plenum. The external leakage resistances at the inlet and the outlet ports of the hydro-motor are modelled by the elements R_6 (R_{lkg1}) and R_{21} (R_{lkg2}), respectively. The cross-port leakage resistance R_{ipl} due to the pressure difference of the hydro-motor is appended at 1 junction. The other two external leakages at the upstream and downstream sides of the system are modelled by R_9 (R_{high}) and R_{22} (R_{low}) elements connected at 0 junctions. The numerical values of these resistances (R_{high} and R_{low}) are toggled from a high to low value or vice versa, according to the switching mode of the DCV. The low value resistance is always connected to the sump. In bond graph (Fig. 3), the transformer element (TF) is used to transform the hydraulic power to mechanical power and vice versa. The D_{m} is the volume displacement rate of the motor and assigned as the modulus of the TF element. The losses due to pipe resistance at upstream and downstream sides are modelled by appending R_{bl1} and R_{bl2} at 1 junction. The viscous resistance (R_{f}) and the moment of inertia (J)

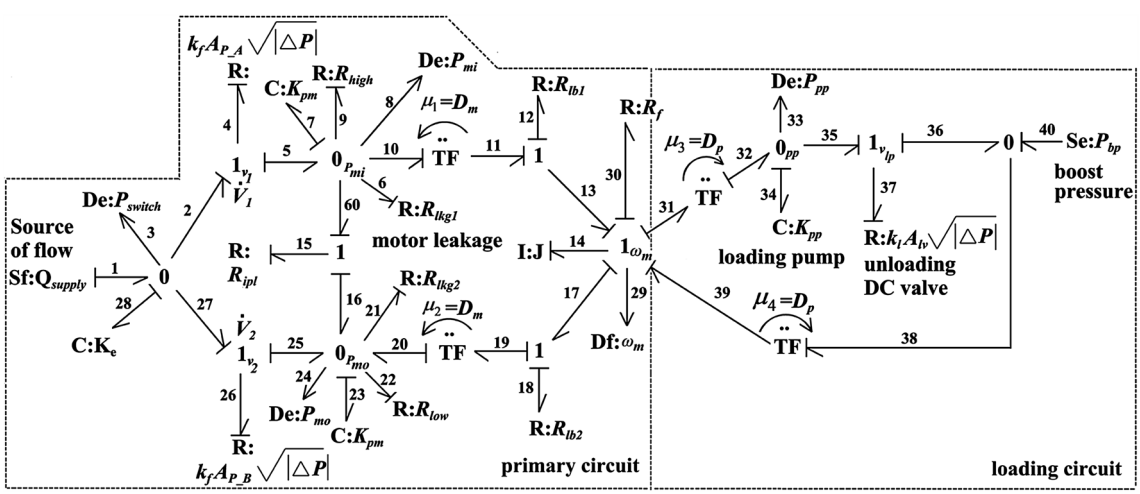

Fig. 3 Bond graph model of the open-loop HST system without soft switch

of the rotating shaft, which determines the angular speed (ω_m), are connected to the 1:ω_m junction. The angular speed of the hydro-motor (ω_m) is measured by a flow detector (Df_{29}).

In the loading circuit, the TF modulus D_p is the volume displacement rate of the radial piston pump with respect to the shaft rotation ω_m. The loading pump plenum pressure P_{pp} is measured by an effort detector at 0 junction (De:P_{pp}). The unloading DC valve (item 14 in Fig. 1) and the pressure relief valve (item 6 in Fig. 1) are installed in parallel for varying the load pressure P_{pp}. The fluid can flow either through the unloading valve or the relief valve depending upon the user command given to the solenoid S_3. Therefore, these two valves are modelled using a single R element, the port opening area (A_{lv}), which can be varied according to the situation.

3.2 Bond graph model of the system with soft switch

This section describes the bond graph model of the soft switch integrated to the basic system model. The integrated model is shown in Fig. 4. The transformer (TF) element, which is connected between 0 and 1_{ss}, converts the pressure energy into mechanical energy. The preload of the spring ($F_{preload}$), stiffness of the spring (K_{ss}), the mass of the soft switch piston (M_{ss}), and the friction between the piston and the soft switch wall (R_{sp}) are represented by Se, C, I and R elements, respectively, and all are connected to 1 junction that constitutes the load dynamics of the soft switch. Another transformer (TF) element, which is connected between 0_{pp} and 1_{ss}, converts the mechanical energy to pressure energy.

The pressure at the rear end of the soft switch (P_{rear}) is measured at 0_{ss} junction. The C element at 0_{ss} junction represents the effective bulk stiffness of the fluid (K_e) at the rear end of the soft switch. The resistance of the flow through the check valve (from the tank to the soft switch) and through the port of the soft switch to the tank are modelled by the R elements at 1_{ch} and 1_{pt} junction, respectively. Similarly, the check valve, which permits the flow from the rear end of the soft switch to the system for regeneration purpose, is modelled by another resistance R at 1_{rg} junction.

The fluid flow (Q) through all the valves, i.e., direction control valve (DCV), pressure relief valves (PRV), check valves, and the soft switch ports, follows a general nonlinear orifice flow equation as:

$$Q = k_f A_v \sqrt{|\Delta P|}\,\mathrm{sign}(\Delta P) \tag{1}$$

where $K_f = C_d\sqrt{2/\rho}$ is the valve flow constant, C_d is the coefficient of discharge, ρ is the mass density of the working fluid, and A_v is the valve orifice area. The sign (..) function (or signum function) is an odd mathematical function that extracts the sign of a real number x given in

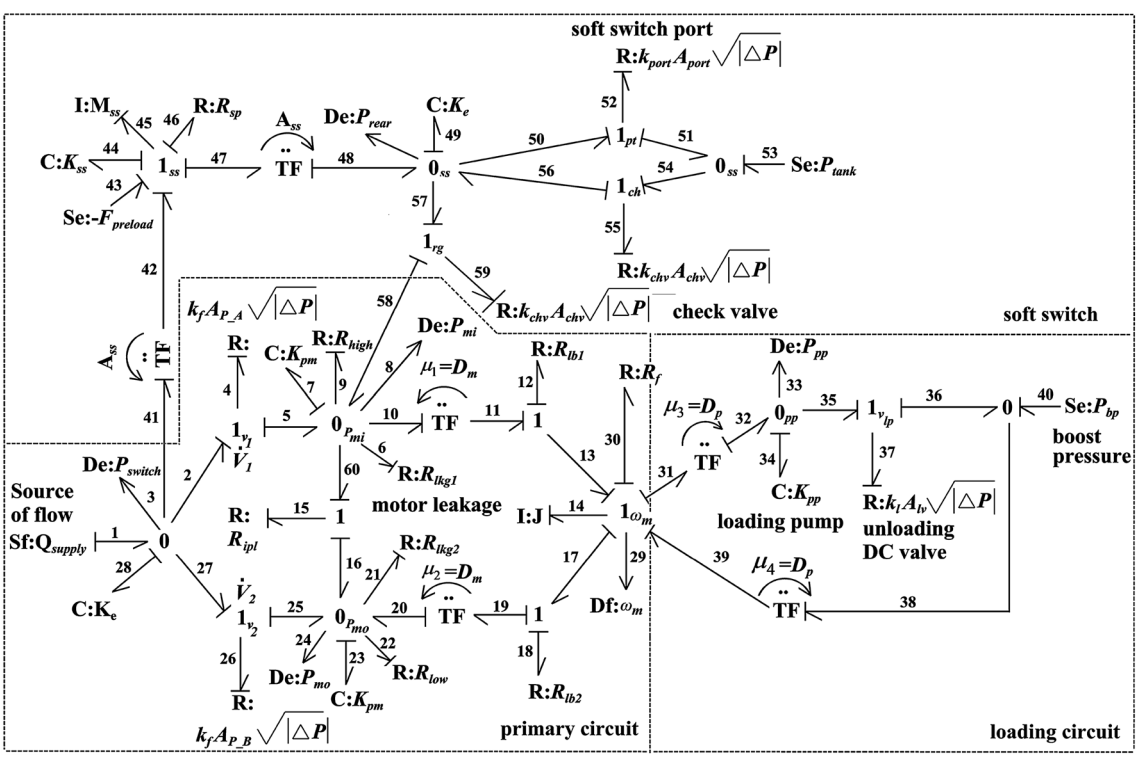

Fig. 4 Bond graph model of the system integrated with the soft switch

its argument. The mathematical expression of the sign (..) function is expressed as:

$$\mathrm{sign}(x) = \begin{cases} -1 & \text{if } (x < 0), \\ 0 & \text{if } (x = 0), \\ 1 & \text{if } (x > 0), \end{cases} \tag{2}$$

The fluid bulk modulus of the working fluid is expressed as $K_e = -V(\mathrm{d}P/\mathrm{d}V)$, which is a function of the fluid pressure. The switched volume pressure (P_{switch}) can be obtained from Eq. (3).

$$\dot{P}_{\mathrm{switch}} = \frac{K_e(P_{\mathrm{switch}})}{V_{\mathrm{switch}} + A_{\mathrm{ss}}x_{\mathrm{ss}}}.\left(Q_{\mathrm{supply}} - Q_{\mathrm{dcvP_x}} - A_{\mathrm{ss}}v_{\mathrm{ss}}\right) \tag{3}$$

In Eq. (3), Q_{supply} is the constant flow supply by the pump and $Q_{\mathrm{dcvP_x}}$ is the flow rate through the 4/2 DCV port. The subscript x is replaced by A and B for the flow through port P-A and P-B, respectively. The A_{ss}, V_{ss} and x_{ss} are the area, the velocity, and the instantaneous position of the piston of the soft switch, respectively. The bulk modulus depends upon different factors, i.e., the working conditions, the amount of entrapped air or gas, and the compression process [23].

The effective bulk modulus (K_e) of fluid is pressure-dependent [23] and it is expressed as:

$$K_e(p) = K\left[\frac{\left(P/P_0\right)^{\left(1/\gamma\right)}\ell^{\frac{(P-P_0)}{K}} + R_a}{\frac{R_aK}{\gamma P} + \left(P/P_0\right)^{\left(1/\gamma\right)}\ell^{\frac{(P-P_0)}{K}}}\right], \tag{4}$$

where K is the bulk modulus of air free flowing fluid, P_0 is the atmospheric pressure, γ (=1.4) and R_a (=0.02) are the ratio of specific heat and volume fraction of air entrained in the oil, respectively. The flow through 4/2 DCV is given in Eq. (5)

$$Q_{\mathrm{dcvP_x}} = k_f A_{\mathrm{P_x}}\sqrt{|(P_{\mathrm{switch}} - P_{\mathrm{mi}})|}\,\mathrm{sign}(P_{\mathrm{switch}} - P_{\mathrm{mi}}) \tag{5}$$

where $A_{\mathrm{P_x}}$ is the port opening area of the 4/2 DCV.

Similarly, the rate of the flow through the unloading DC valve (item 14 in Fig. 1) or the relief valve (item 6 in Fig. 1) depends on the port opening area of the orifice (A_{lv}) and the pressure difference $\Delta P = (P_{\mathrm{pp}} - P_{\mathrm{bp}})$ across it, and it is expressed as:

$$Q_{\mathrm{unv}} = R_{37} : Sf_{37} = k_l A_{\mathrm{lv}}\sqrt{|(P_{\mathrm{pp}} - P_{\mathrm{bp}})|}\,\mathrm{sign}(P_{\mathrm{pp}} - P_{\mathrm{bp}}), \tag{6}$$

where k_l is the flow constant.

The velocity of the soft switch piston [13] is estimated from Eq. (7) using the Newton's second law.

$$\dot{v}_{\mathrm{ss}} = \frac{(P_{\mathrm{switch}} - P_{\mathrm{rear}})A_{\mathrm{ss}} - F_{\mathrm{preload}} - K_{\mathrm{ss}}x_{\mathrm{ss}} - \frac{2\pi\mu v_{\mathrm{ss}}r_{\mathrm{ss}}l_{\mathrm{ss}}}{C_{\mathrm{ss}}}}{M_{\mathrm{ss}}} \tag{7}$$

In Eq. (7), the r_{ss} and l_{ss} are the radius and length of the piston of the soft switch, respectively. The dynamic viscosity of the flowing fluid and the bore clearance of the piston are represented by μ and C_{ss}, respectively. The pressure at the rear end of the soft switch piston (P_{rear}) is obtained from Eq. (8)

$$\dot{P}_{\mathrm{rear}} = \frac{K_e(P_{\mathrm{rear}})}{V_{\mathrm{ss_clear}} + A_{\mathrm{ss}}(x_{\mathrm{ss_max}} - x_{\mathrm{ss}})} \times (A_{\mathrm{ss}}v_{\mathrm{ss}} + Q_{\mathrm{chv_tb}} - Q_{\mathrm{chv_ip}} - Q_{\mathrm{ss_port}}), \tag{8}$$

where $V_{\mathrm{ss_clear}}$ is the clearance volume at the rear end of the soft switch when it is at maximum travel, x_{ss_max} is the maximum travel length. $Q_{\mathrm{chv_tb}}$ and $Q_{\mathrm{chv_ip}}$ are the rate of the flow through the two check valves from the tank to the rear end of the soft switch and from the soft switch to the mainstream, respectively. $Q_{\mathrm{ss_port}}$ is the rate of the flow through the port of the soft switch to the tank. These flow rates as assigned in the bond graph model in Fig. 4 are:

$$Q_{\mathrm{chv_tb}} = \begin{cases} 0, & \text{if } P_{\mathrm{crack}} > (P_{\mathrm{tank}} - P_{\mathrm{rear}}) \\ k_{\mathrm{check}}\sqrt{|(P_{\mathrm{tank}} - P_{\mathrm{rear}} - P_{\mathrm{crack}})|}\,\mathrm{sign}(P_{\mathrm{tank}} - P_{\mathrm{rear}} - P_{\mathrm{crack}}), & \text{if } P_{\mathrm{crack}} \le (P_{\mathrm{tank}} - P_{\mathrm{rear}}) \end{cases} \tag{9}$$

$$Q_{\mathrm{chv_ip}} = \begin{cases} 0, & \text{if } P_{\mathrm{crack}} > (P_{\mathrm{rear}} - P_{\mathrm{mi}}) \\ k_{\mathrm{check}}\sqrt{|(P_{\mathrm{rear}} - P_{\mathrm{mi}} - P_{\mathrm{crack}})|}\,\mathrm{sign}(P_{\mathrm{rear}} - P_{\mathrm{mi}} - P_{\mathrm{crack}}), & \text{if } P_{\mathrm{crack}} \ge (P_{\mathrm{rear}} - P_{\mathrm{mi}}) \end{cases} \tag{10}$$

$$Q_{\mathrm{ss_port}} = K_{\mathrm{port}}A_{\mathrm{port}}\sqrt{|(P_{\mathrm{rear}} - P_{\mathrm{tank}})|}\,\mathrm{sign}(P_{\mathrm{rear}} - P_{\mathrm{tank}}) \tag{11}$$

In Eqs. (9–11), P_{crack} is the cracking pressure of the check valve, $k_{check}(=C_D A_{check}\sqrt{2/\rho})$ is the flow constant of the check valve, C_D is the coefficient of the flow, A_{check} is the port opening area of the orifice of the check valve, A_{port} is the port opening area of the port of the soft switch, and K_{port} is the flow constant of the port of the soft switch. The check valves are considered to open instantly. The conditional opening of the port of the soft switch is described as given in Eq. (12).

$$A_{port} = \begin{cases} 0, & \text{if } (x_{ss} < x_{ss_crack}) \\ 2\pi r_{ss}(x_{ss} - x_{ss_crack}), & \text{if } (x_{ss_crack} \le x_{ss} < x_{ss_crack} + l_{ss_port}) \\ 2\pi r_{ss} l_{ss_port}, & \text{if } (x_{ss} \ge x_{ss_crack} + l_{ss_port}) \end{cases} \quad (12)$$

where l_{ss_port} is the axial length of the port of the soft switch.

The equations of state of the system are found from the bond graph model (Fig. 4), which are given in Eqs. (13–19).

$$Q_{supl} - A_{ss}v_{ss} - k_f A_{P_A}\sqrt{|P_{switch} - P_{mi}|}\,\text{sign}(P_{switch} - P_{mi}) - k_f A_{P_B}\sqrt{|P_{switch} - P_{mo}|}\,\text{sign}(P_{switch} - P_{mo}) - \frac{\dot{P}_{switch}}{K_e} = 0 \quad (13)$$

$$k_f A_{P_A}\sqrt{|P_{switch} - P_{mi}|}\,\text{sign}(P_{switch} - P_{mi}) - D_m\omega_m - \frac{P_{mi}}{R_{lkg1}} - \frac{P_{mi} - P_{mo}}{R_{ipl}} - \frac{\dot{P}_{mi}}{K_{pm}} - \frac{P_{mi}}{R_{high}} + k_{check}\sqrt{|(P_{rear} - P_{mi} - P_{crack})|}\,\text{sign}(P_{rear} - P_{mi} - P_{crack}) = 0 \quad (14)$$

$$k_f A_{P_B}\sqrt{|P_{switch} - P_{mo}|}\,\text{sign}(P_{switch} - P_{mo}) + D_m\omega_m - \frac{P_{mo}}{R_{lkg2}} + \frac{P_{mi} - P_{mo}}{R_{ipl}} - \frac{\dot{P}_{mo}}{K_{pm}} - \frac{P_{mo}}{R_{low}} = 0 \quad (15)$$

$$D_p\omega_m - k_l A_{lv}\sqrt{|P_{pp} - P_{bp}|}\,\text{sign}(P_{pp} - P_{bp}) - \frac{\dot{P}_{pp}}{K_{pp}} = 0 \quad (16)$$

$$D_m(P_{mi} - P_{mo}) - \omega_m(R_{bl1} + R_{bl2} + R_f) - D_p(P_{pp} - P_{bp}) - J\dot{\omega}_m = 0 \quad (17)$$

$$A_{ss}v_{ss} - \frac{P_{rear}}{K_e} + k_{check}\sqrt{|P_{tank} - P_{rear} - P_{crack}|}\,\text{sign}(P_{tank} - P_{rear} - P_{crack}) - k_{port}A_{port}\sqrt{|P_{rear} - P_{tank}|}\,\text{sign}(P_{rear} - P_{tank}) - k_{check}\sqrt{|P_{rear} - P_{mi} - P_{crack}|}\,\text{sign}(P_{rear} - P_{mi} - P_{crack}) = 0 \quad (18)$$

$$(P_{switch} - P_{rear})A_{ss} - F_{preload} - K_{ss}x_{ss} - R_{sp}v_{ss} - M_{ss}\dot{v}_{ss} = 0 \quad (19)$$

Note that if the soft switch terms (A_{ss}, x_{ss}, v_{ss}, R_{ss}, K_{ss}, M_{ss}, p_{rear}, $F_{preload}$) and the check valves terms (k_{check}, P_{crack}, P_{tank}) are removed from the equations of state (Eqs. 13–19), they result in governing equations for the system model without the soft switch (Fig. 3).

4 Results and discussions

4.1 Multi-run simulation

In Sect. 3, the system equations (Eqs. 13–19) have been derived from the bond graph model using an established step by step algorithm. The total number of first order differential equations generated from the system is equal to the number of lumped storage elements (i.e., I and C) with integral causality. The state variables of the system in terms derived equations are the absorbed causes in storage elements with integral causalities. Generally, the system variables for bond graph based analysis are generalized momentum, $p = \int e dt$, associated with I element in integral causality and generalized charge/displacement, $q = \int f dt$, associated with C element in integral causality, where e and f denote generalized effort and flow, respectively. The system equations are generated by writing down the element constitutive relations and the junction and two-port element constraint equations [24, 25]. The software Symbols-Shakti [26] adopts the bond graph algorithm for generation of the system equations (ODE) in its graphical model builder.

The equations are then solved in a different module, simulation workbench, which uses different integration methods like Runge–Kutta Gill, Runge–Kutta–Fehlberg, Gear's Predictor-Corrector, etc. Among the abovementioned methods, the Runge–Kutta Gill has been adopted to integrate the equations of state for the present application. This is a fifth-order Runge–Kutta method, which propagates a solution over an interval by combining the information from several Euler-style steps (each involving one evaluation of the equations of state), and then using the information obtained to match a Taylor series expansion. The differential equations are solved as an initial value problem. The prior behavior of a converged solution is not used, only the end states of a solution are used as the initial state of the next simulation step. This is mathematically proper, since any point along the trajectory in state-space of a deterministic system can serve as an initial point. For convergence, the step size in time stepping is dynamically adjusted so that the relative truncation error is less than a small value, which is chosen as 5×10^{-6} for the present simulation study. The parameter values given in Table 2 are used for the model simulation.

Table 2 List of the parameters used in simulation

Description	Value
Volume displacement rate of motor, D_m	1.3057×10^{-6} m^3 rad^{-1}
Volume displacement rate of pump, D_p	0.85×10^{-6} m^3 rad^{-1}
Constant supply flow, Q_{supply}	25×10^{-5} m^3 s^{-1}
Loading pump plenum pressure, P_{bp}	6.5 bar
External leakage resistance, ($R_{lkg} = R_{lkg1} = R_{lkg2}$)	6.5×10^{18} N s m^{-5}
Cross-port leakage resistance, R_{ipl}	17×10^{18} N s m^{-5}
Viscous friction resistance, R_f	1.6×10^{-5} N s m rad^{-1}
Pipe friction resistance, ($R_{lb} = R_{lb1} = R_{lb2}$)	1×10^{-2} N s m rad^{-1}
DCV flow constant, k_f	4.811×10^{-2} kg$^{-0.5}$ m$^{1.5}$
Load inertia, J	1×10^{-3} kg m^2
Mass density of the oil, ρ	870 kg m^{-3}
Port opening area of the DCV (A_{P_A})	30×10^{-6} m^2
Port opening area of the DCV (A_{P_B})	15×10^{-6} m^2
Bulk modulus of fluid, K	1.9×10^9 N m^{-2}
Bulk stiffness of fluid ($K_{pm} = K_{pp}$)	1×10^{14} N m^{-5}
Switched volume, V_{switch}	1×10^{-6} m^3
Unloading DC valve flow constant, k_l	4.79×10^{-2} kg$^{-0.5}$ m$^{1.5}$
Port opening area of the unloading DC valve, A_{lv}	3.5×10^{-6} m^2
Dynamic viscosity of oil, μ	1 N s m^{-2}
Tank pressure, P_{tank}	1.013 bar
Piston radius, r_{ss}	4.5×10^{-3} m
Length of piston, l_{ss}	13×10^{-3} m
Clearance volume at the rear end of piston, V_{ss_clear}	1×10^{-6} m^3
Radial clearance of piston, C_{ss}	10 µm
Check valve cracking pressure, P_{crack}	550 bar
Flow constant of check valve, k_{check}	7.375×10^{-6} kg$^{-0.5}$ m$^{3.5}$

The bond graph model of the hydraulic system integrated with soft switch has been simulated using optimized parameters of the soft switch, obtained by multiple run simulation. The critical parameters taken into consideration for optimization with respect to minimum throttling energy losses are the spring stiffness (K_{ss}), the preloading force of the spring ($F_{preload}$), the crack length of the piston travel (x_{ss_crack}) and the maximum possible travel length (x_{ss_max}). Multiple run simulations are carried out by varying the parameter values independently within a certain realistic bound (K_{ss}: 1×10^6–5×10^4 N m^{-1}, $F_{preload}$: 10–19 N, x_{ss_crack}: 1×10^{-3}–9×10^{-3} m, x_{ss_max} : 10×10^{-3}–17.5×10^{-3} m) and specifying the mode of interpolation. The mode of interpolation has two scales, i.e., *linear map* and *logarithmic map*. For the present work, *logarithmic map* has been used, and therefore, both the starting and ending values have the same sign. The results obtained therefrom have been examined and compared for finding out the parameter values corresponding to the minimum throttling energy loss.

Table 3 Regression data

S. no.	x_{ss_crack}	K_{ss}	$F_{preload}$	x_{ss_max}	$E_{throtloss}$
1	0.001	1,000,000	10	0.01	1075.37
2	0.001123	854,131	10.3436	0.010299	1051.92
3	0.00126	729,541	10.699	0.010607	1032.23
4	0.001415	623,124	11.0666	0.010924	1013.98
5	0.001588	532,230	11.4468	0.01125	999.903
6	0.001783	454,594	11.8401	0.011587	982.754
7	0.002001	388,283	12.2469	0.011933	969.415
8	0.002247	331,645	12.6677	0.01229	952.09
9	0.002522	283,268	13.103	0.012657	941.355
10	0.002831	241,948	13.5532	0.013035	931.671
11	0.003179	206,656	14.0189	0.013425	919.118
12	0.003568	176,511	14.5005	0.013826	911.079
13	0.004006	150,764	14.9987	0.01424	903.529
14	0.004497	128,772	15.5141	0.014665	906.506
15	0.005048	109,988	16.0471	0.015104	894.793
16	0.005667	93,944.4	16.5985	0.015555	885.524
17	0.006362	80,240.9	17.1688	0.01602	874.668
18	0.0078	58,539	18.3689	0.016992	863.774
19	**0.009**	**50,000**	**19**	**0.0175**	**847.971**

During multiple run simulation, when the spring stiffness is reduced from 1×10^6 to 5×10^4 N m^{-1}, x_{ss_crack} increases and throttling energy loss reduces (decreases) or vice versa. If the spring stiffness is larger than 1×10^6 N m^{-1}, the x_{ss_crack} value is lower than 1 mm, which is difficult to machine. On the contrary, when the spring stiffness is lower than 5×10^4 N m^{-1}, residual error is developed during the locking of the soft switch. Similarly, increasing the spring preloading force, x_{ss_crack} decreases or vice versa. The large preload force on the spring is avoided, because it decreases the value of x_{ss_crack}, thereby demanding more precision manufacturing. On the other hand, lower value of preload force produces residual error during locking of the soft switch. The regression data generated from multi run simulation are given in Table 3.

Optimization denotes improving the performance of a system to obtain the maximum advantage out of it. Conventionally, optimization is carried out by monitoring the effect of one factor at a time on an experimental response. While only one parameter is changed, others are kept at a constant level. This optimization technique is called one-variable-at-a-time. Its major shortcoming is that it does not include the interactive effects among the variables studied. As a consequence, this method does not illustrate the complete effects of the parameter on the response [27]. Another drawback of

the one-factor optimization is the increase in the number of experiments necessary to conduct the research, which leads to an increase of time and expenses.

To overcome these problems, the optimization is carried out using multivariate statistic techniques. Among the multivariate techniques used in analytical optimization, response surface methodology (RSM) is most popular [28, 29]. Response surface methodology is a collection of mathematical and statistical techniques based on the fit of a polynomial equation to the experimental data, which must depict the behavior of a data set with the objective of making statistical perception. This can be useful when a response or a set of responses of interest are influenced by several variables. The objective is to simultaneously optimize the levels of these variables to achieve the best system performance.

In this present work, a first-degree polynomial model has been used and the parameters (K_{ss}, x_{ss_crack}, $F_{preload}$, x_{ss_max},) have been changed simultaneously. An easy way to estimate a first-degree polynomial model is to use a factorial experiment or a fractional factorial design, which has been applied through Minitab software in the present analysis.

If the parameter values of Table 3 are varied independently to achieve the minimum throttling energy loss ($E_{throtloss}$), the corresponding x_{ss_crack} values must be changed accordingly, because its value is dependent to other parameters (K_{ss}, $F_{preload}$, x_{ss_max}). That means, if K_{ss} value is changed from 1×10^6 to 854,131 N m^{-1} independently, then the crack length of the piston travel (x_{ss_crack}) must be increased from 0.001 m to *some new value* to open the port of the soft switch because with the reduction of K_{ss} the x_{ss_crack} value is increased. However, if x_{ss_crack} is considered 0.001 m in the second row in the above table, then the port of the soft switch and the piston annulus cut-out will coincide earlier and the flow will be passed through the port of the soft switch before completion of the cycle. Hence, the unlocking process of the soft switch will be initiated earlier resulting in undesired loss of pressurized fluid to the tank. Therefore, the function of the soft switch in view of energy savings will become meaningless.

Similarly, reversed situation will occur. If the value of $F_{preload}$ is increased from 10 to 10.3436 N, then the x_{ss_crack} value must be decreased from 0.001 m to *another value*. In such cases, the port of the soft switch and the piston annulus cut-out will not coincide, as a result the soft switch will not able to release the developed pressure and it does not work properly.

In Table 3, the row (Sl. No. 19) corresponding to the minimum throttling energy loss ($E_{throtloss}$) is highlighted in bold font. The soft switch parameter values (i.e., x_{ss_crack}, K_{ss}, $F_{preload}$ and x_{ss_max}), as given in Sl. No. 19, have been assigned in the model simulation. A mathematical relationship has been developed from the data obtained

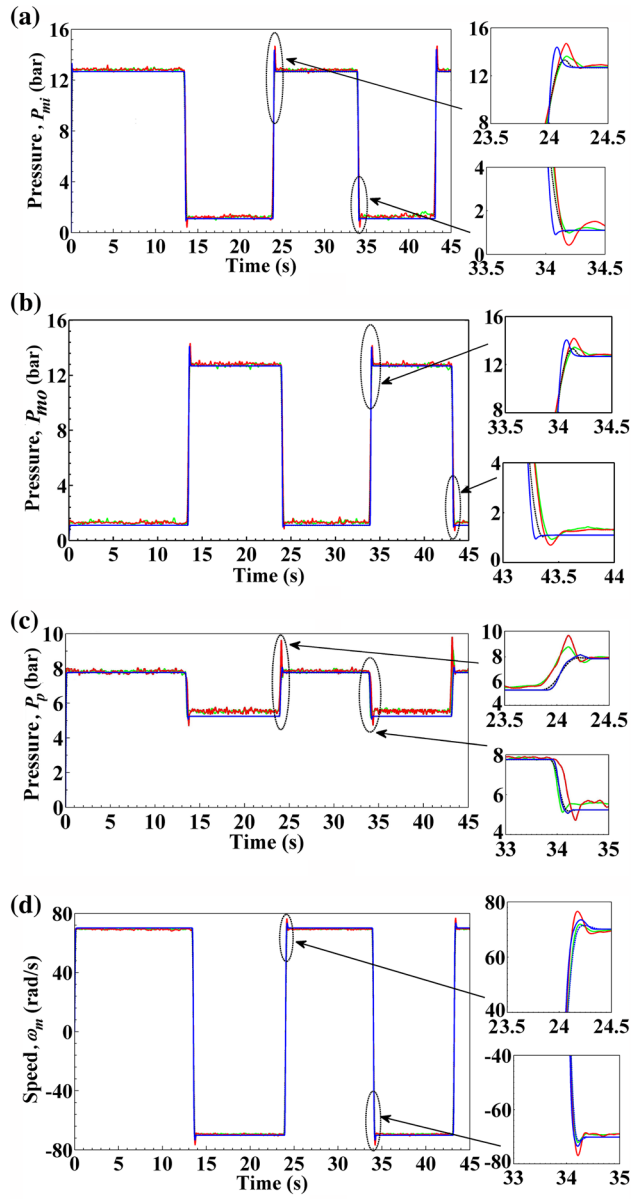

Fig. 5 Comparison of the state responses of the system *without* and *with* soft switch: **a** pressures at the motor inlet (P_{mi}), **b** Pressure at the motor outlet (P_{mo}), **c** pressure at the pump plenum (P_p) and **d** angular speed of the motor (ω_m) (color figure online)

by multi-run-simulation (Table 3), using multiple regression analysis among the critical parameters and throttling energy loss, discussed in the next subsection.

4.2 Regression analysis

The statistical tool which is used to examine how multiple independent variables are related to a dependent variable is called multiple regression analysis. Regression generally uses the ordinary least square method which derives

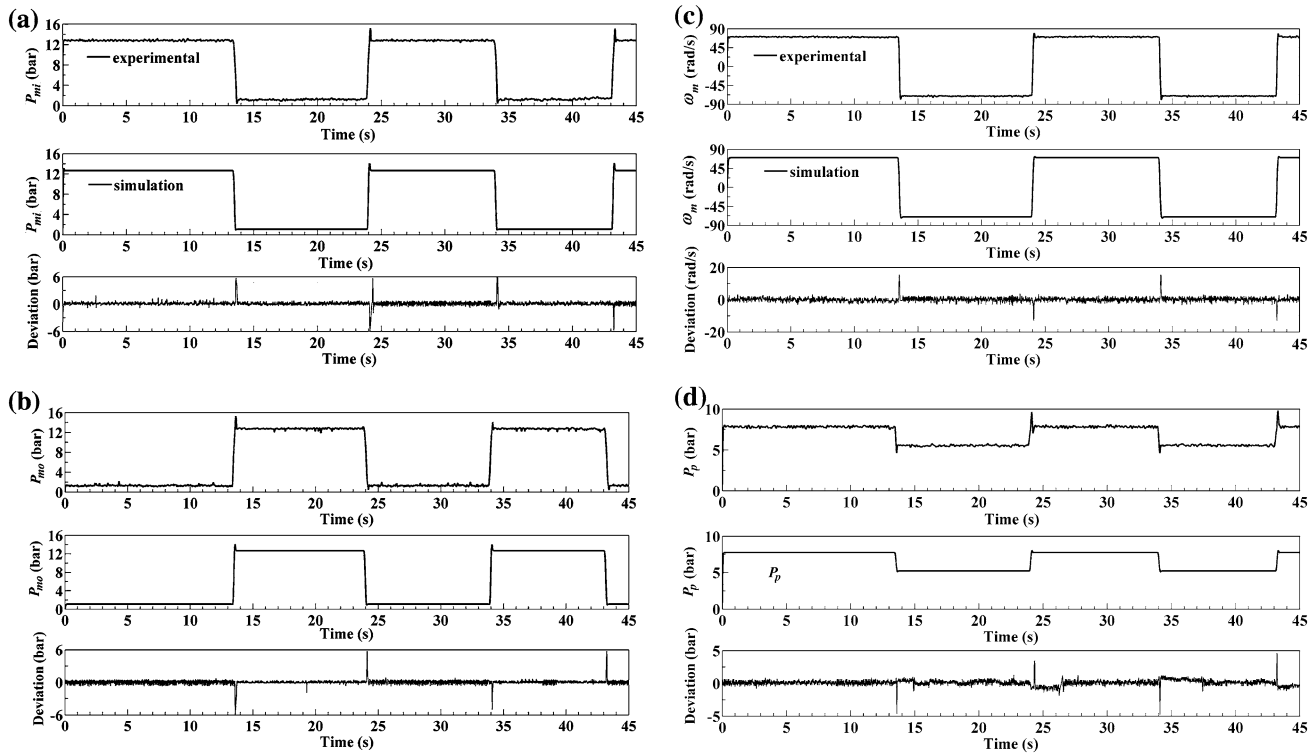

Fig. 6 State responses and errors corresponding to 7.7 bar load pressure. **a** Comparison of experimental and simulation results of P_{mi}, **b** comparison of experimental and simulation results of P_{mo}, **c** comparison of experimental and simulation results of ω_m, **d** comparison of experimental and simulation results of P_p

the equation by minimizing the sum of the squared residuals (the distances between the sample's data points and the values predicted by the equation) in the given dataset. The regression model for estimating the throttling energy loss with known values of key parameters (i.e., x_{ss_crack}, K_{ss}, $F_{preload}$ and x_{ss_max}) is given in Eq. (20).

$$E_{throtloss} = a_0 + a_1 x_{ss_crack} + a_2 K_{ss} + a_3 F_{preload} + a_4 x_{ss_max} \tag{20}$$

where, a_i ($i = 0 \ldots 0.4$) are constant coefficients and are obtained using Minitab software. Here, a_0 implies the intercept point and other terms (a_1, a_2, a_3, and a_4) are the slopes. The constant coefficients are obtained through least square minimization so as to obtain a best fit.

$$E_{throtloss} = 2377.77 - 37791.65\, x_{ss_crack} + 9.12 \times 10^{-6} K_{ss} + 784.5\, F_{preload} - 919966\, x_{ss_max} \tag{21}$$

The value of the regression coefficient (R^2) is found to be 0.997, which is near to unity. Therefore, the regression model is acceptable from the accuracy viewpoint.

Equation (21) can be used to estimate the throttling energy loss ($E_{throtloss}$) of a system for a given set of parameters (x_{ss_crack}, K_{ss}, $F_{preload}$ and x_{ss_max}) of the soft switch avoiding simulation or experimentation. Alternatively, for attaining a desired value of $E_{throtloss}$ a critical parameter (for example x_{ss_crack}) can be computed when all other parameters (i.e., K_{ss}, $F_{preload}$, x_{ss_max}) are known. By assigning the optimized parameters of the soft switch from Table 3 (Sl. item 19) in Eq. (21), the estimated throttling energy loss is obtained as 844.046 J, which is close to the value of energy loss as given in Table 3 (847.971 J).

4.3 Experimental validation

The pressure and the angular speed responses from the system model (without and with soft switch) are validated with test data archived in a portable data acquisition system, HMG 3010 [30]. The HMG 3010 is designed primarily for recording measured values of the variables of pressure, flow rate and angular speed that are common in hydraulics. The measurement ranges and units are automatically detected by the HMG 3010. The ability to connect the HMG 3010 to a CAN (Controller Area Network) bus and thus to display messages from the CAN bus expands the range of applications. It can be logically linked to PC connection through USB–RS 232 port. Up to 10 sensors can be connected simultaneously and up to 32 measurement channels can be displayed at

Table 4 Statistical data calculated during 3–7 s when P-A port of the DCV was open

Experiment no.	Condition	Measured state	Mean	Median	Mode
I	Flow was passed through unloading DC valve at *lower* loading pressure	P_{mi}	12.81 bar	12.82 bar	12.75 bar
		P_{mo}	1.289 bar	1.28 bar	1.139 bar
		ω_m	70.29 rad s^{-1}	70.3 rad s^{-1}	70.21 rad s^{-1}
		P_p	7.853 bar	7.851 bar	7.844 bar
II	Flow was passed through unloading DC valve (experiment *I* repeated)	P_{mi}	12.92 bar	12.91 bar	12.43 bar
		P_{mo}	1.403 bar	1.384 bar	1.196 bar
		ω_m	70.4 rad s^{-1}	70.37 rad s^{-1}	69.47 rad s^{-1}
		P_p	7.824 bar	7.82 bar	7.742 bar
III	Flow was passed through relief valve at *higher* loading pressure	P_{mi}	13.36 bar	13.37 bar	12.81 bar
		P_{mo}	1.319 bar	1.32 bar	1.12 bar
		ω_m	64.57 rad s^{-1}	64.48 rad s^{-1}	63.38 rad s^{-1}
		P_p	9.923 bar	9.923 bar	9.908 bar
IV	Flow was passed through relief valve (experiment III repeated)	P_{mi}	13.26 bar	13.24 bar	13.19 bar
		P_{mo}	1.329 bar	1.311 bar	1.102 bar
		ω_m	65.37 rad s^{-1}	65.28 rad s^{-1}	64.13 rad s^{-1}
		P_p	9.937 bar	9.81 bar	9.907 bar

a time. Measuring rates can be set up to 0.1 ms and voltage measurement range is −10 to +10 V, which can be extended from 0 to 50 V.

The test data for the present experiment (i.e., P_{mi}, P_{mo}, P_p and ω_m) were recorded at a sampling time of 5 ms as shown in Fig. 5. The results were obtained by switching on/off the directional solenoid valve (DCV). During the abrupt closing and opening of the DCV ports, the pressures and the angular speed overshot significantly, as observed in Fig. 5. The red and the blue color curves of Fig. 5 indicate the experimental and simulation responses of the system states when the soft switch is not connected, respectively, whereas the green and the black color curves denote the same with the soft switching.

When the DCV starts to switching from its open port position (P-A) to its cross port position (P-B), the pressure at the inlet of the motor (P_{mi}) decreases from 12.67 to 1.099 bar and the pressure at the downstream of the motor (P_{mo}) increases from 1.099 to 12.67 bar. The pressure at the pump plenum (P_{pp}) changes from 7.76 to 5.24 bar. The angular speed of the hydraulic motor (ω_m) changes from 70.14 to −70.14 rad s^{-1}, i.e., reverses its direction of rotation. Similarly, when the DCV is switched from its cross port position (P-B) to open port position (P-A), the pressures and the angular speed revert back to their original values. These processes continue in the next cycle. The steady state values of the responses in Fig. 5 are almost the same in both *with* and *without* soft switch system. The only significant changes are observed during the valve transitions. We observe that the transient pick during the valve spool transition is lower when the soft switch is integrated to the system (green and black color curves) as compared to the system without soft switch (red and blue color curves). Hence, it can be concluded that the soft switch helps to smoothen the transition operations and contributes to energy savings to some extent.

To examine the repeatability of the experiment before recording data through the data logger [27], four number of experiments were conducted at different times with the same load pressure setting and also different setting ($P_p = 7.7$ and 9.8 bar). Figure 6 shows the results obtained by conducting the experiment at lower pressure (passing the fluid through the unloading DC valve, item 14 in Fig. 1). The deviation between the test data and the simulation results of all the measured states are observed mostly during switching that are shown as spikes in the deviation plots. This happens because of actuation response delay in experiments. At other places, the deviation is bounded and limited to ±5%.

In the next experiment, power supply to the solenoid S_3 of the bypass valve (item 14 in Fig. 1) was switched off, thus forcing the fluid to flow through the relief valve (item 06 in Fig. 1) at a higher set pressure ($P_p = 9.8$ bar). Thereafter, two experiments were conducted at the same setting but at different times and the statistical data of all the experiments are compared in Tables 4 and 5, which show the consistency of results in repeated experimental run. The deviations of all the measured states are computed again for this experiment (i.e., $P_p = 9.8$ bar, although graphs are not shown for the sake of brevity), which are mostly consistent within the deviation limit of ±5%, except during a short period after switching. This short period deviations (shown as spikes in deviation plots) occur due to the

Table 5 Statistical data calculated during 14–18 s when P-B port of the DCV was open

Experiment no.	Condition	Measured state	Mean	Median	Mode
I	Flow was passed through unloading DC valve at *lower* loading pressure	P_{mi}	1.185 bar	1.185 bar	0.9 bar
		P_{mo}	12.86 bar	12.74 bar	12.45 bar
		ω_m	−70.17 rad s^{-1}	−70.21 rad s^{-1}	−70.28 rad s^{-1}
		P_p	5.509 bar	5.51 bar	5.422 bar
II	Flow was passed through unloading DC valve (experiment *I* repeated)	P_{mi}	1.352 bar	1.335 bar	1.247 bar
		P_{mo}	12.81 bar	12.81 bar	12.69 bar
		ω_m	−69.65 rad s^{-1}	−69.59 rad s^{-1}	−70.25 rad s^{-1}
		P_p	5.408 bar	5.363 bar	5.253 bar
III	Flow was passed through relief valve at *higher* loading pressure	P_{mi}	1.251 bar	1.277 bar	1.233 bar
		P_{mo}	13.45 bar	13.44 bar	13.4 bar
		ω_m	−65.95 rad s^{-1}	−65.88 rad s^{-1}	−66.11 rad s^{-1}
		P_p	3.259 bar	3.259 bar	3.241 bar
IV	Flow was passed through relief valve (experiment III repeated)	P_{mi}	1.279 bar	1.292 bar	1.233 bar
		P_{mo}	13.48 bar	13.45 bar	13.23 bar
		ω_m	−66.13 rad s^{-1}	−66.02 rad s^{-1}	−66.17 rad s^{-1}
		P_p	3.258 bar	3.237 bar	3.233 bar

actuator response delay in the experiments. The internal dynamics of the solenoid valve has not been modelled in this article and hence the delay is not found in the simulation results; thereby giving short periods of response mismatch. Hence, it can be concluded that all the experimental responses are within the acceptable margin.

4.4 Uncertainty analysis

The process of evaluating the deviation associated with a measurement is often called uncertainty analysis or sometimes error analysis. The uncertainty estimate associated with a measurement should account for both the accuracy and precision of the measurement. Accuracy assumes that there is a standard or known value, and expresses how far the measured value is deviated from that. Precision, on the other hand, refers to the closeness of two or more measurements to each other. The uncertainty of a result is usually expressed as the root-sum-square (RSS) of the bias (i.e., accuracy) and precision limits [31, 32]. This present study is focused on uncertainty in estimating the throttling power loss ($P_{throtloss}$).

The throttling power loss of the system (refer Fig. 5) is:

$$P_{throtloss} = (P_{switch} - P_{mi})Q_{dcvP_A} + (P_{switch} - P_{mo})Q_{dcvP_B} \quad (22)$$

Substituting the expression of Q_{dcvP_x} (note that the subscript x is replaced by A and B for flow through port P-A and P-B, respectively) from Eq. (3) into Eq. (22),

$$\begin{aligned} P_{throtloss} = {} & P_{switch}\left[P_{mi}\left(\frac{1}{R_{lkg}} + \frac{1}{R_{high}}\right) + P_{mo}\left(\frac{1}{R_{lkg}} + \frac{1}{R_{low}}\right)\right] \\ & - P_{mi}\left[D_m\omega_m + P_{mi}\left(\frac{1}{R_{lkg}} + \frac{1}{R_{high}}\right) + \frac{P_{mi} - P_{mo}}{R_{ipl}}\right] \\ & - P_{mo}\left[P_{mo}\left(\frac{1}{R_{lkg}} + \frac{1}{R_{low}}\right) - \frac{P_{mi} - P_{mo}}{R_{ipl}} - D_m\omega_m\right]. \end{aligned} \quad (23)$$

The throttling power loss ($P_{throtloss}$) of the system depends on four measured entities: P_{switch}, P_{mi}, P_{mo} and ω_m. Therefore, $P_{throtloss} = f(P_{switch}, P_{mi}, P_{mo}, \omega_m)$.

The total experimental uncertainty ($U_{Pthrotloss}$) related to the estimation of throttling power loss is [31]:

$$U_{Pthrotloss} = \sqrt{\left\{(S_{Pthrotloss})^2 + (P_{Pthrotloss})^2\right\}}, \quad (24)$$

Table 6 Uncertainty analysis of throttling power loss

Variables (r)	No. of samples (M)	Standard deviation $SD = \sqrt{\frac{\sum(x_i - \bar{x})^2}{M-1}}$	Systematic error (S_i)	$\frac{\partial P_{throtloss}}{\partial r}$	Precision error (P_i)
P_{switch}	1000	0.141978 bar	5×10^{-6} bar	2.49958×10^{-4} m^3 s^{-1}	8.97×10^{-3} bar
P_{mi}	1000	0.133551 bar	5×10^{-6} bar	-2.4128×10^{-4} m^3 s^{-1}	8.45×10^{-3} bar
P_{mo}	1000	0.107987 bar	5×10^{-6} bar	0.1022×10^{-4} m^3 s^{-1}	6.83×10^{-3} bar
ω_m	1000	2.881339 rad s^{-1}	0.01 rad s^{-1}	−1.51 N m rad^{-1}	0.18223 rad s^{-1}

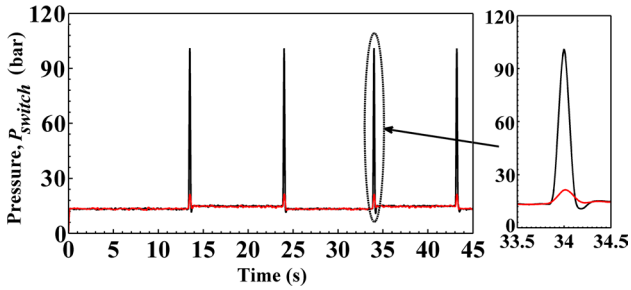

Fig. 7 Comparison of the switched volume pressure

where $S_{\text{Pthrotloss}}$ and $P_{\text{Pthrotloss}}$ are the systematic limit and the precision limit of the throttling power loss, respectively, which are expressed as:

$$(S_{\text{Pthrotloss}})^2 = \left(\frac{\partial P_{\text{throtloss}}}{\partial P_{\text{switch}}} S_{\text{Pswitch}}\right)^2 + \left(\frac{\partial P_{\text{throtloss}}}{\partial P_{\text{mi}}} S_{\text{Pmi}}\right)^2 + \left(\frac{\partial P_{\text{throtloss}}}{\partial P_{\text{mo}}} S_{\text{Pmo}}\right)^2 + \left(\frac{\partial P_{\text{throtloss}}}{\partial \omega_{\text{m}}} S_{\omega m}\right)^2 \tag{25}$$

$$(P_{\text{Pthrotloss}})^2 = \left(\frac{\partial P_{\text{throtloss}}}{\partial P_{\text{switch}}} P_{\text{Pswitch}}\right)^2 + \left(\frac{\partial P_{\text{throtloss}}}{\partial P_{\text{mi}}} P_{\text{Pmi}}\right)^2 + \left(\frac{\partial P_{\text{throtloss}}}{\partial P_{\text{mo}}} P_{\text{Pmo}}\right)^2 + \left(\frac{\partial P_{\text{throtloss}}}{\partial \omega_{\text{m}}} P_{\omega m}\right)^2 \tag{26}$$

In Eqs. (25–26), S_{Pswitch}, S_{Pmi}, S_{Pmo}, $S_{\omega\text{m}}$ and P_{Pswitch}, P_{Pmi}, P_{Pmo}, $P_{\omega\text{m}}$ are the bias and precision error of the variables P_{switch}, P_{mi}, P_{mo} and ω_{m}, respectively. The bias error of each variable (B_i) is provided by the manufacturer; whereas the precision error $\left(P_i = \frac{t_{\text{factor}} \times \text{standard deviation(SD)}_i}{\sqrt{\text{sample number }(M)}}\right)$ of each variable are estimated from the archived data. The t_{factor} is the coverage factor and t_{factor} is 2 for $M \geq 30$ for 95% level of confidence [29]. The uncertainties of the four measured state variables are computed on the data during the time period from 1 to 6 s (when steady state is attained) and given in Table 6.

Using the numerical values of systematic and precision errors (as in Table 6), in Eqs. (25–26), $(S_{\text{Pthrotloss}})^2$ and $(P_{\text{Pthrotloss}})^2$ are estimated as 2.28×10^{-4} and 0.16768 W, respectively. Substituting those values in Eq. (24), the total experimental uncertainty error ($U_{\text{Pthrotloss}}$) is obtained as ± 0.41 W $\approx 4\%$ with respect to the mean value of throttling power loss (9.42 W) during the time period from 1 to 6 s.

4.5 Comparison between both systems: without soft switch and with soft switch

The variations of the switched volume pressure for both systems are shown in Fig. 7. The maximum value of switched pressure has been decreased from 101 to 21.5 bar due to the introduction of the soft switch in the system. The black color plot denotes the switch pressure responses of the system when the soft switch is not connected to the system and the red one indicates the smoothening of the same pressure during the transition when the soft switch is integrated.

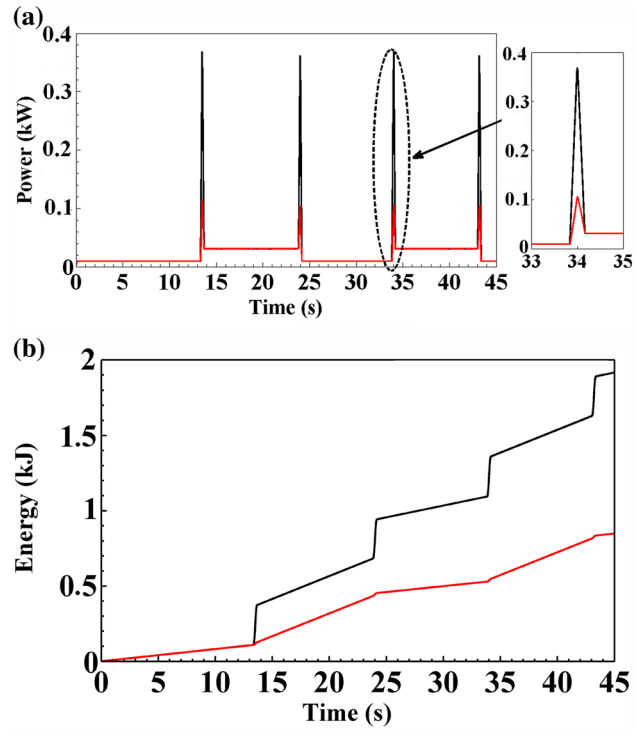

Fig. 8 Comparison of throttling losses in 4/2 DCV **a** throttling power loss, **b** throttling energy loss

The resistive element (R) is a power dissipating component, which means the power [i.e., product of effort (e) and flow (f) variables associated with this element] is lost from the system in the form of heat or some other form of energy. From Fig. 3, the power loss during the throttling through the DCV can be written as:

$$P_{\text{throtloss}}(t) = e_4 f_4 + e_{26} f_{26} \tag{27}$$

The energy loss can be obtained by taking the time integral of the power loss, i.e.,

$$E_{\text{throtloss}} = \int P_{\text{throtloss}} \mathrm{d}t \tag{28}$$

The simulated power loss (kW) and the throttling energy loss (kJ) for both the systems are compared in Fig. 8a, b, respectively. The maximum power loss decreases from 0.37 to 0.18 kW by incorporation the soft switching. Similarly, the throttling energy loss is reduced from 1.92 to 0.85 kJ (i.e., 56% savings). This quantification of energy savings cannot

be verified through experimentation as the flow rate through the DCV is not measurable with the available instrumentation. However, the throttling power and energy loss of the system (with and without soft switch) during the valve transition are quantified through simulation of the validated model. The black color curves show the variation of the power loss and the energy loss of the system without soft switch, while the red curves represent the same quantity when the soft switch is connected to the system.

5 Conclusion

The experimental investigation of the performance of an open-loop hydrostatic drive system with the inclusion of soft switch has been presented with respect to throttling energy saving. The throttling energy loss is quantified through the analysis. It has been revealed that the throttling energy saving of the DCV with soft switch is 56%, as compared with the system when the soft switch is not connected.

The check valve (item 13) that is used for safety purpose in the soft switch circuit is directly connected with the upstream side of the hydraulic circuit. But, with the spool shifting of the DCV from the open port to cross port or vice versa, the upstream and downstream line to the motor will also be interchanged, leading to the speed reversal of the motor. Accordingly, the connection from the rear end of the soft switch piston of the pressure line should be altered to accomplish the full benefit of regeneration. An appropriate design for that may be taken up in the future.

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